# Jeppe × Jakob 1-1: NaturArv3D + other stuff — 2026-08-04

# Overview of Time Spent

- Mostly compression study
- Some `qim3d` work
  - Would like to do a bit of a sprint before many new people start, but need Felipe/Jakob
- Not insignificant time wrangling HPC
  - Despite HPC TA, need better ways of interacting with HPC
  - Some time spent on prototype tool `ezhpcy`
  - Some time spent on `dask-jobqueue` setup

# Qim3D

- Mostly little fixes for bugs I found when using `qim3d` for compression study
- Top priority:
  - `pre-commit`/formatting currently creates a lot of friction
  - tests *need* to be run in CI
    - 46 fail, 180 pass
    - Currently this creates a lot of extra work during review

# Compression Study — Overview

- WIP: Report (can demo)
- Have gotten very nice structure set up for reproducible experiments
  - Care has gone into eliminating IO
    - We leverage `/dev/shm` as RAM-disk
    - Some variability tests
  - Ensuring consistent HPC environment
  - …
- Gotten overview of most relevant compression methods/codecs
  - Is TIFF hard requirement?
  - File sizes?

# Compression Study — Preliminary Literature Review

- Mostly medical imaging, relatively few with projection data, mostly just reconstructions
- [1], [2]
  - $\min(\mathrm{SSIM}_{\mathrm{local}}) < 0.99$ indistinguishable by radiologists
  - Lossless has best-case compression ratios $\approx 1.70 : 1$ (JPEG-LS) to $\approx 1.95 : 1$ (proprietary from paper)
  - Lossy compression about $3 : 1$ while indistinguishable for radiologists (we can do better?)
- Some reading up on `imagecodecs` and `zarr`

# Compression Study — Preliminary Findings

- JPEG-XL seems very promising
  - Recently DICOM standard [3]
  - Best-in-class for lossless
  - Works for `u8`, `u16`, `f16`, `f32`
- LERC promising because guaranteed error bounds
  - Can propagate under linear assumptions?
  - Works for just about any data type

# Compression Study — Codec Overview (WIP)

| Encoding | Pixel Types | Channels | Type | Categories | Unsuitable Reason |
|---|---|---|---|---|---|
| **JPEG 1 (lossy)** | **u8**, **u12** (mono) | **1/3** | L | DICOM, TIFF? | Bit-Depth |
| **JPEG 1 (lossless)** | **u1-16** | **1/3** | LL | DICOM, TIFF? | Bit-Depth? |
| **Run Length Encoding (RLE, DICOM)** | **u1/8/16**, u1-64 | **1/3** | LL | DICOM, TIFF | |
| **JPEG-LS** | **u2-16**, u2-32 | **1/3** | L, LL | DICOM | |
| **JPEG 2000** | **u1-38** | **1/3**, $\leq 16384$ | L, LL | DICOM, TIFF? | |
| **JPEG 2000 (HTJ2K)** | **u1-38** | **1/3**, $\leq 16384$ | L, LL | DICOM | |
| **JPEG-XL** | **u1-24**, u1-32 | **1/3**, $\leq 4099$ | L, LL | DICOM, TIFF? | |
| WEBP | u8 | 3/4 | L, LL | DICOM, TIFF? | Bit-Depth |
| AVIF | u8/10/12 | $\leq 2^{32}+1$ | L, LL | | Bit-Depth |
| LERC | i/u8/16/32, f32/64 | $\leq 2^{31}-1$ | L, LL | TIFF? | |
| Blosc | sd | | | | |
| Zstd | | | ddd | | |
| LZW | Any | N/A | LL | TIFF | |
| ZFP | | | | | |
| SZ3 | (i/u8/16/32/64), f32/64 | | L, (LL) | | |
| CCITT Group 3 (G3) | u1 | N/A | LL | TIFF | Bit-Depth |
| CCITT Group 4 (G4) | u1 | N/A | LL | TIFF | Bit-Depth |
| SGILOG | u8/16, f32 | 1/3 | LL | | |

Table 1: **Bold** refers to configurations supported by the DICOM specification. *TIFF?* refers to codecs that have been granted a TIFF compression tag, but are not widely supported by TIFF readers, whereas *TIFF* refers to codecs that are widely supported by TIFF readers. Note that TIFF is a container format and can contain codecs that have not been granted a TIFF compression tag. If the *Unsuitable Reason* column is not empty, the codec has been deemed unsuitable for inclusion in this study on the basis of the reason provided.

# Compression Study — Experiments

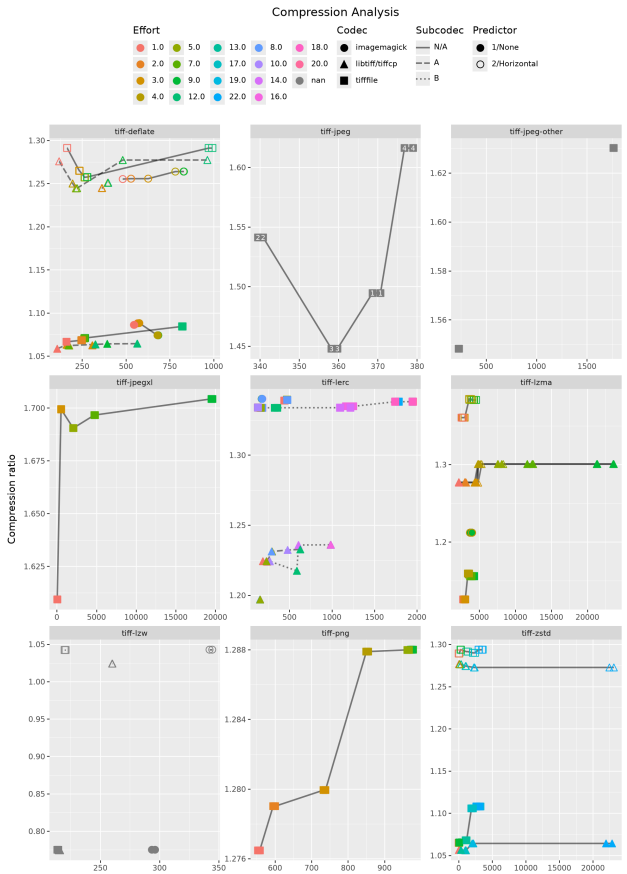
- Main experiments:
  - WIP: Lossless, tiff-stack, projections compression
  - WIP: Lossless, zarr, projections compression
  - WIP: Lossy, tiff-stack, reconstructions compression
  - WIP: Lossy, zarr, reconstructions compression
  - (lossy reconstructions -> projections study)
- Smaller experiments:
  - Mostly done: Variability in encoding/decoding speeds
  - WIP/Not started: File size scaling
    - WIP: zarr chunking initial study

Note: Some Zarr stuff, though not shown here.

- I much prefer Zarr to TIFF!

# Compression Study — Lossless TIFF Stack Compression

- WIP and made for report, so quite tiny
- JPEG-XL seems best
  - 1.7 CR => 40% space savings (for free!)
  - Very new format, still improvements to be made
    - Found some critical bugs bugs, hopefully fixed in `0.13` (September?)
    - Will improve: adopted in PDF and DICOM standards
- JPEG-LS also good
- LERC somewhat lack-luster, unfortunately
- Standard bytestream codecs (zstd, lzma, deflate, lzw) are pretty bad



# Compression Study — Lossy JPEG-XL TIFF Reconstruction Compression

- Distance $d = 0.1$, `f32` data ($1000 \times 1000 \times 1000$ shape):
  - $99.55\%$ of pixels have SSIM $> 0.99$
  - Compression ratio of $12.9$ ($92.2\%$ space savings!)
- Hopefully ran lower distance tests over night, not sure if finished
- Pretty comparison picture missing :(

# Show Code

- Maybe quickly show code any over-all structure

# Bibliography

[1] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and A. Et., "Effects of Fixed-Rate CT Projection Data Compression on Perceived and Measured CT Image Quality," *SPIE Proceedings*, 2010, doi: 10.1117/12.841145.

[2] A. Wegener, N. Chandra, Y. Ling, R. Senzig, and R. Herfkens, "Real-Time Compression of Raw Computed Tomography Data: Technology, Architecture, and Benefits," presented at the SPIE Medical Imaging, E. Samei and J. Hsieh, Eds., Lake Buena Vista, FL, Feb. 2009, p. 72582H. doi: 10.1117/12.810599.

[3] "DICOM PS3.5 2026c — Data Structures and Encoding." Accessed: Jul. 17, 2026. [Online]. Available: https://www.dicomstandard.org/current